MATH 2211, SPRING 2026: MIDTERM 1

This exam has 6 (six) printed pages. Please show all your work. Outside materials are NOT permitted for use on this test. Do not spend more than 10 minutes on any one part of a problem. If you get stuck for more than a few minutes, just move on to the next part. *You should feel free to use the results of earlier problems in your solutions to later ones, regardless of whether you have finished them or not.* You have 50 minutes to work on the test. Write your answers in the space provided.

Name ___________________________________

	Score
Problem 1	/10
Problem 2	/10
Problem 3	/10
Problem 4	/10
Problem 5	/10
Total	/50

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(1) (10 points) **True or false. Write one or two lines justifying your answer**:

(a) If F is a field and $a \in F$, then $(-1) \cdot a = -a$.

(b) $\mathbb{Z}/6\mathbb{Z}$ with modular addition and multiplication is a field.

(c) There exists a set of 5 linearly independent vectors inside $\mathbb{R}^4$.

(d) The equation $z^n = 3i + 4$ admits n distinct solutions for z in $\mathbb{C}$.

(e) The polynomials $1 + x, 1 - x \in \text{Poly}(\mathbb{Z}/3\mathbb{Z})$ are linearly independent.

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(2) (10 points) Show that, for any integer $n \geq 1$, we have

$$1 \cdot 1! + 2 \cdot 2! + \cdots + n \cdot n! = (n+1)! - 1$$

Here, $n!$ is is *n-factorial*, defined recursively by $1! = 1$ and $n! = n \cdot (n-1)!$, so that, for $n \geq 2$, we have $n! = n(n-1) \cdots 2 \cdot 1$.

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(3) (10 points) Consider the vectors

$$v_1 = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}, v_2 = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}, v_3 = \begin{bmatrix} 0 \\ -1 \\ 3 \end{bmatrix}, v_4 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \in \mathbb{Q}^3$$

(a) Find all solutions for $x_1, x_2, x_3, x_4 \in \mathbb{Q}$ to the equation

$$x_1 v_1 + x_2 v_2 + x_3 v_3 + x_4 v_4 = \begin{bmatrix} 3 \\ -2 \\ 0 \end{bmatrix}$$

(b) Can you find a subset of $\{v_1, v_2, v_3, v_4\}$ that is a basis for $\mathbb{Q}^3$?

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(4) (10 points) Consider the subset

$$U = \{f(x) \in \mathrm{Poly}_{\leq 3}(\mathbb{R}) : \int_0^1 f(x)\, dx = 0\}.$$

(a) Show that U is an $\mathbb{R}$-subspace of $\mathrm{Poly}_{\leq 3}(\mathbb{R})$.

(b) Find a basis for U and compute $\dim(U)$.

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(5) (10 points) Give a rigorous proof of the following assertion: If V is a finite dimensional F-vector space and $W \subset V$ is an F-subspace with $\dim W = \dim V$, then in fact $W = V$.